Week 2

Topic: Divide and Conquer

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324. Wiggle Sort II (Medium)

Given an integer array nums, reorder it such that nums[0] < nums[1] > nums[2] < nums[3]....

You may assume the input array always has a valid answer.

Example 1:

Input: nums = [1,5,1,1,6,4]

Output: [1,6,1,5,1,4]

Explanation: [1,4,1,5,1,6] is also accepted.

Example 2:

Input: nums = [1,3,2,2,3,1]

Output: [2,3,1,3,1,2]

For detailed descriptions and constraints, please refer to Leetcode 324.

493. Reverse Pairs (Hard)

Given an integer array nums, return *the number of **reverse pairs** in the array*.

A **reverse pair** is a pair (i, j) where:

- 0 <= i < j < nums.length and
- nums[i] > 2 * nums[j].

Example 1:

Input: nums = [1,3,2,3,1]

Output: 2

Explanation: The reverse pairs are:

(1, 4) --> nums[1] = 3, nums[4] = 1, 3 > 2 * 1

(3, 4) --> nums[3] = 3, nums[4] = 1, 3 > 2 * 1

Example 2:

Input: nums = [2,4,3,5,1]

Output: 3

Explanation: The reverse pairs are:

(1, 4) --> nums[1] = 4, nums[4] = 1, 4 > 2 * 1

(2, 4) --> nums[2] = 3, nums[4] = 1, 3 > 2 * 1

(3, 4) --> nums[3] = 5, nums[4] = 1, 5 > 2 * 1

For detailed descriptions and constraints, please refer to Leetcode 493.

23. Merge k Sorted Lists (Hard)

You are given an array of k linked-lists lists, each linked-list is sorted in ascending order.

Merge all the linked-lists into one sorted linked-list and return it.

Example 1:

Input: lists = [[1,4,5],[1,3,4],[2,6]]

Output: [1,1,2,3,4,4,5,6]

Explanation: The linked-lists are:

[

1->4->5,

1->3->4,

2->6

]

merging them into one sorted linked list:

1->1->2->3->4->4->5->6

For detailed descriptions and constraints, please refer to Leetcode 23.